INSTRUCTOR GUIDE

Technology, Space & Society

Course 01 — Facilitation Guide for Graduate Assistants and Faculty

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HOW TO USE THIS GUIDE

This guide is designed for three audiences: (1) graduate teaching assistants facilitating discussion sections on behalf of the course instructor; (2) faculty at other institutions who wish to adopt or adapt this course; and (3) Patrick Hoffman himself, as a scaffold for replicating the course across different institutional contexts.

Each session entry contains:

- A Session Overview describing the intellectual purpose of the class and what students should leave with
- A Theoretical Frame orienting the facilitator to the key arguments in the readings
- A Suggested Discussion Arc with approximate timing
- Numbered Discussion Questions (1–3) for all students, with facilitation notes on each
- Graduate-level questions (G1–G2) marked with a teal accent, for master's and doctoral students
- Common Misconceptions — the errors students reliably make, and how to address them
- Connections to other sessions — how this session's arguments thread through the course
- Facilitator Notes — logistical and pedagogical guidance specific to this session

A note on Patrick's voice in this course

This course is grounded in Patrick Hoffman's professional arc — from Columbia architecture and urban planning, through building Social Bicycles (the product that became JUMP), to product leadership at Google and Meta. That biography is not incidental to the course; it is a teaching resource. When @arcktip readings or case studies appear, facilitators should encourage students to interrogate the practitioner perspective alongside the academic sources — not to privilege one over the other, but to ask what each can see that the other cannot.

On adapting for your own institution

The course was designed for a dual-listed seminar (upper-division undergraduate + master's) in an environment that brings together students from architecture, urban planning, and media or STS programs. If your institution runs these populations separately, the graduated question structure (numbered questions for all, G-marked questions for graduate students) makes separation clean. If you are teaching a purely undergraduate or purely graduate course, adjust reading loads accordingly and use the reading list document to calibrate.

Week 1 · Introduction: Technology Redefines Space

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

The first session sets the intellectual stakes of the entire course. Students arrive with intuitions about how technology 'changes things,' but rarely with a framework for thinking about spatial and social consequences specifically. The goal is to surface those intuitions, complicate them, and install the course's central question: not just what does a technology do, but what does it do to the spaces and social arrangements that surround it?

THEORETICAL FRAME

Winner's 'Do Artifacts Have Politics?' (1980) is the conceptual anchor. The argument — that technologies embed social relations and authority structures — is simple enough to grasp on a first reading, but rich enough to sustain a full seminar. The Moses case (raised highways as racial barriers) is vivid and contestable, making it ideal for opening discussion. Petroski's evolutionary framing offers a gentler entry point: things look the way they do because of decisions made by people.

SUGGESTED DISCUSSION ARC

0–10 min	Instructor opens with a single image: a 1950s American living room with a television set, furniture rotated toward it. No caption. Ask: 'What happened here, and why?' Collect observations on the board.
10–25 min	Brief lecture: introduce the course question — 'How do technologies reshape the spaces we inhabit?' Frame the three levels of analysis: the room, the city, the platform. Preview the 15-week arc.
25–50 min	Seminar discussion on Winner. Open with Question 1, work through Questions 2–3. Pause at minute 40 to introduce Petroski's 'form follows failure' idea as a counterpoint.
50–65 min	Graduate students: push into Question G1–G2. Undergrads: redirect to a contemporary case (the iPhone's home screen as spatial reorganization of attention).
65–75 min	Close with 'the assignment': students must identify one technology in their daily life that has changed a space. They'll introduce it Week 2 in 60 seconds.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Langdon Winner argues that the Long Island parkway overpasses were built low to keep buses — and therefore poor and Black New Yorkers — away from Jones Beach. Does it**

matter whether the historical claim is literally true? What's the argument even if it's not?
Let students debate the historical accuracy first. Then pivot: the form of the argument is what matters. Can you think of a contemporary artifact with analogous embedded politics?

2 **Petroski says that the shape of everyday objects is determined by the failures of prior versions. How does that change how you look at, say, the layout of a smartphone's home screen? Or a city block?**
Goal: students start seeing designed objects as accumulated decisions, not natural facts.

3 **What's the difference between a technology that changes what you can do, and one that changes where you are while you do it? Can you find an example of each?**
Bridge question — starts building the spatial vocabulary we'll use all semester.

GRADUATE / MASTER'S QUESTIONS

G1 MacKenzie and Wajcman argue that technology is 'socially shaped' — not socially determined, but not autonomous either. How does that sit alongside Winner's claim that artifacts have politics? Is Winner a social constructivist?

G2 Mumford distinguishes between the 'eotechnic,' 'paleotechnic,' and 'neotechnic' phases of technological development. Does that periodization still hold? Where would you slot the smartphone?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often read Winner as saying technology is always political. He's making a narrower claim: some technologies have political consequences baked into their design. Sharpen the distinction.

Many students arrive with a naive technological determinism — tech changes society. The course argument is more reciprocal. Push back gently in Week 1 so you have room to build.

Graduate students sometimes over-theorize immediately. Encourage them to stay close to a concrete object or space before reaching for ANT or SCOT.

CONNECTIONS TO OTHER SESSIONS

This session's core question — 'what does a technology do to a space?' — should be referenced explicitly every time a new case is introduced. It is the thread. In Week 15, return to the living room image from today and ask what students would say now.

FACILITATOR NOTES

> Write the course question on the board and leave it there: 'How do technologies reshape the spaces we inhabit?' Keep returning to it. For classes with a mix of architecture, planning, and media students, the living room image tends to read differently across disciplines — use that productively.

Week 2 · The Hearth → The Television

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

Spigel's Make Room for TV is the course's first sustained case study. The argument — that the introduction of television into postwar American homes required a massive cultural negotiation about gender, domesticity, and family life, not just a furniture rearrangement — models the kind of analysis students will produce themselves. This session rewards close reading: Spigel's sources (women's magazine advertisements, TV criticism, interior design guides) are unusual and generative.

THEORETICAL FRAME

Spigel works in the tradition of cultural studies and feminist media history. She is not asking what television did to people; she is asking how people made sense of television's arrival, and how that sense-making shaped what television became. The gap between the two questions is the heart of the session.

SUGGESTED DISCUSSION ARC

0–10 min	Students present their 60-second 'technology in my daily life' observations from last week's assignment. Take 4–5. Note commonalities on the board.
10–20 min	Ask: 'Before there was a TV, where was the center of the American living room? How do you know?' Brief discussion. Then show a progression: fireplace → piano → radio cabinet → television set → laptop. What's constant? What changes?
20–45 min	Core seminar on Spigel. Use Questions 1–3. Focus especially on Question 2 — the gender dynamics are where students tend to have the richest disagreements.
45–60 min	Graduate section: push into Morley (ethnographic method) and Silverstone (domestication theory). Question G1 and G2.
60–75 min	Close: 'What's the equivalent of the television in your home right now? What furniture did you rearrange for it? What did that rearrangement cost you?' Brief reflection, no wrong answers.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Spigel argues that television wasn't simply adopted into the home — it had to be 'made room for,' culturally as well as architecturally. What's the difference? Why does it**

matter?
Students often flatten this. Push: the furniture arrangement was easy. What was hard?

2 **In the magazine sources Spigel analyzes, television is simultaneously celebrated as a tool that brings family together and blamed for disrupting domestic order. Who bears the burden of managing that contradiction in the texts she reads?**
This is where gender enters. Don't lead with it — let students find it in the texts.

3 **Spigel's sources are women's magazines and TV criticism — not traditional historical archives. What does that methodological choice enable? What does it foreclose?**
Good bridge to the graduate discussion on Morley's ethnographic approach.

GRADUATE / MASTER'S QUESTIONS

G1 Morley's Family Television (1986) uses ethnographic interviews to study how families actually watch TV, not how they're told to. What does the ethnographic method reveal that Spigel's discourse analysis cannot? Are they asking the same question?

G2 Silverstone's domestication theory describes four phases: appropriation, objectification, incorporation, conversion. Map the television's entry into the postwar home onto those four phases using Spigel's evidence. Where does the model strain?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read Spigel as nostalgic or prescriptive — as if she thinks we should have kept the fireplace. She's describing a negotiation, not lamenting an outcome. Keep the analysis sociological, not moral.

The gender argument can feel dated to undergraduates. Reframe: 'Who manages technology anxiety in your household right now? Who sets up the router, updates the software, troubleshoots the smart home?' The pattern often persists.

Graduate students may want to jump to ANT immediately. Hold them in Spigel for the first half — close reading first, theory second.

CONNECTIONS TO OTHER SESSIONS

Return to Week 1's core question. Connect to Week 5 (telephone and threshold — another communication technology that required domestic negotiation) and Week 11 (smart speakers — the contemporary version of this same story). The 'hearth sequence' (fireplace → piano → TV → laptop → smart speaker) is a throughline for the whole course.

FACILITATOR NOTES

> The 60-second student presentations can run long. Set a visible timer. The goal is not full analysis — just: 'Here's my technology, here's the space it changed.' You'll use their examples all semester.

Week 3 · STS Frameworks

SESSION FORMAT	TIMING	LEVEL
Seminar (theory-heavy)	90 min	UG + Grad

SESSION OVERVIEW

This is the hardest session for undergraduates and the most important for graduate students. Actor-Network Theory, SCOT, and domestication theory are the analytical tools students will use for the rest of the course. The goal is not comprehensive mastery — it's enough fluency to deploy these frameworks on a case. Latour in particular requires patience. Budget time for confusion and work through it together rather than papering over it.

THEORETICAL FRAME

Three frameworks are in play: SCOT (Pinch & Bijker) argues that technology's meaning is constructed through the interpretive flexibility of relevant social groups. ANT (Latour, Callon) goes further — nonhuman actors (machines, scallops, door-closers) have agency too, and must be enrolled in networks. Domestication theory (Silverstone) focuses on the household as the site where technologies are tamed into everyday life. These are not the same argument. The session should make the distinctions clear.

SUGGESTED DISCUSSION ARC

0–15 min	Start with a simple object: a bicycle. Ask students to describe it using each framework in turn. SCOT: who are the relevant social groups, and what does the bicycle mean to each? ANT: what actors need to be enrolled for a bicycle to work? Domestication: how does a bicycle get incorporated into a household routine? Work through all three quickly — this is a warm-up, not the main event.
15–40 min	Seminar on Winner and Pinch/Bijker: how does SCOT differ from Winner's artifacts-have-politics argument? Question 1. Then shift to Latour — the hotel key example, the door-closer. Question 2.
40–60 min	Break into groups of 3–4. Each group picks one technology from the course so far (TV, car, PC) and maps it onto one of the three frameworks. Share back in plenary. Instructor synthesizes.
60–80 min	Graduate section: Callon's scallops paper. Question G1. Then Haraway — Question G2.
80–90 min	Close: 'Which framework do you find most generative for your research proposal topic? Write one sentence.' Collect — use to give proposal feedback.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Pinch and Bijker argue that the 'safety bicycle' wasn't obviously safer or better — it won because of interpretive flexibility across different social groups. Does that mean there's no such thing as a better technology? Or a worse one?**

This question usually produces strong disagreement. Let it run. The goal is to surface the difference between technical and social criteria of success.

2 **Latour asks us to treat the hotel key fob (a heavy weight attached to the key so guests return it) as an actor. What does it mean for an object to have a 'program of action'? Does the fob have interests?**

Keep this concrete. Resist the urge to make it immediately metaphysical.

3 **All three frameworks — SCOT, ANT, domestication — treat technology as embedded in social relations. But they emphasize different things: meaning, networks, households. For the technology you're researching this semester, which lens seems most productive, and why?**

Personal application. Gets students thinking about their own projects.

GRADUATE / MASTER'S QUESTIONS

G1 Callon's fishermen and scallops paper describes 'translation' as a process of enrolling allies in a network. Who or what plays the role of the scallops in Spigel's account of television? Who plays the fishermen?

G2 Haraway's 'situated knowledges' essay challenges the idea of a 'view from nowhere' in academic analysis. How should an STS researcher's positionality shape their analysis of a technology? Does Patrick Hoffman's practitioner background make his account of dockless bikes more or less reliable?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students frequently confuse ANT's 'network' with a telecommunications network or a social network. Latour's network is a web of associations, human and nonhuman, that must be actively maintained. The word 'actor-network' doesn't refer to the internet.

SCOT is not relativism — it doesn't say all interpretations are equally valid or that the 'better' technology never wins. It says the criteria of 'better' are socially negotiated, not technically given.

Graduate students sometimes use Latour as a way to avoid taking positions. Remind them: ANT is a descriptive method, not an excuse for analytic neutrality.

CONNECTIONS TO OTHER SESSIONS

These frameworks are the analytical toolkit for the whole course. Explicitly return to them whenever a new case arrives. In Week 10 (micromobility), ask: who are the relevant social groups for the dockless scooter? In Week 11 (smart home), ask: what nonhuman actors are enrolled in the Alexa network? Keep the vocabulary alive.

FACILITATOR NOTES

> The bicycle warm-up works extremely well with mixed disciplinary groups — architects, planners, and media scholars all bring different instincts. Grad seminars: Callon's scallops paper often generates productive confusion about what counts as agency. Don't resolve it too quickly — sit with it.

Week 4 ·

The Automobile & the City

SESSION FORMAT	TIMING	LEVEL
Seminar + map exercise	75 min	UG + Grad

SESSION OVERVIEW

The automobile is the most consequential technology of the 20th century for urban form. This session makes that argument concrete by tracing the specific policy mechanisms — zoning codes, FHA mortgage standards, highway funding formulas, parking minimums — through which the car remade the American city. The goal is for students to see urban form not as natural or inevitable, but as the product of specific decisions made by specific people at specific moments.

THEORETICAL FRAME

Hayden and Jackson work as historians, giving students the long view. Shoup's parking minimum argument is the policy lever — it's quantitative, contestable, and directly relevant to contemporary urban debates. Norton's 'Fighting Traffic' adds the forgotten detail that streets were not always conceived as automobile space: the redefinition of the street as a place for cars (not pedestrians, children, or vendors) required an active political campaign by auto interests.

SUGGESTED DISCUSSION ARC

0–10 min	Open with two aerial photographs side by side: downtown Detroit, 1920 vs. 1970. No labels. 'What happened? What's missing? What's new?' Surface: parking lots, highways, missing density.
10–15 min	Brief lecture: the three policy instruments that made suburbia — FHA mortgage standards (racial covenants), the Interstate Highway System, and parking minimums. 5 minutes, no more.
15–45 min	Seminar discussion. Questions 1–3. The Shoup question (Question 2) usually produces the liveliest debate — keep it focused on mechanism, not just critique.
45–60 min	Map exercise: show students a zoning map of any American city. Ask: 'Where can you walk to get food, work, and rest without a car? Mark it.' Discuss what the map reveals about embedded assumptions.
60–75 min	Graduate section: Norton and Fishman. Question G1 and G2. Close with: 'What contemporary mobility technology is doing to cities what the car did in 1920?'

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Hayden argues that suburban development wasn't just a housing preference — it was a policy choice, subsidized by the federal government. Who paid for suburbia, and who was excluded from it?**

Race and the FHA is central here. Don't let students treat this as a settled historical matter — ask whether those spatial patterns persist.

2 **Shoup argues that parking minimums — requirements that every new building include a certain number of parking spaces — have fundamentally distorted American cities. Walk me through the mechanism: how does a parking minimum produce sprawl?**

This is a causal chain question. Push students to trace each step.

3 **If the car was the dominant technology that shaped 20th-century urban form, what technology is doing the same work right now? What spatial patterns will it leave behind?**

Forward-looking. Good for connecting to Weeks 9–12.

GRADUATE / MASTER'S QUESTIONS

G1 Norton's 'Fighting Traffic' argues that the definition of a 'street' as automobile space was actively constructed through lobbying, propaganda, and legal campaigns — it was not obvious or inevitable. Using his framework, describe the analogous campaign being waged today over who owns the curb.

G2 Fishman argues that suburbs represent a specific utopian vision of bourgeois family life. Does that utopia still have purchase? Who holds it now, and has the technology changed?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often think suburban sprawl was the result of consumer preference — people simply wanted houses with yards. Jackson and Hayden show this preference was constructed through policy and subsidy. The demand was shaped before it was expressed.

The parking minimum critique can feel like an abstraction. Make it concrete: find the parking requirement for a building in the city where the course is taught. Ask students to calculate how much land is being used for car storage.

Graduate students sometimes treat this as merely historical. Ask: what is the current equivalent of the 1956 Highway Act? (Possible answers: the infrastructure bill, autonomous vehicle regulations, scooter permit ordinances.)

CONNECTIONS TO OTHER SESSIONS

The automobile session establishes the pattern the course will repeat: a technology arrives, policy decisions lock in spatial consequences, those consequences persist for decades. Connect explicitly to Week 9 (platform urbanism — Uber and the curb) and Week 10 (micromobility — the political fight over street space). In Week 14, return to the question: can cities undo what the car did?

FACILITATOR NOTES

> The map exercise is the most effective teaching tool in this session. Use Google Maps satellite view projected on screen if no printed maps are available. Zoning maps are publicly available for almost every American city. The contrast between a walkable urban grid and a suburban cul-de-sac development is immediately legible.

Week 5 · The Telephone & the Threshold

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

The telephone gets less architectural attention than the car or the TV, but its spatial consequences are profound: it collapsed the distinction between inside and outside, extended the domestic interior into the city and vice versa, and created new kinds of privacy violation and connection. Fischer's social history is the anchor — his account of how telephone companies discovered (and then constructed) the 'social use' of the phone is a model of the kind of analysis this course asks for.

THEORETICAL FRAME

Goffman's front-stage/backstage framework is introduced here as a spatial metaphor, not just a sociological one. The telephone blurs the backstage: when the phone rings in your kitchen, the public world enters your private space. That blurring is the mechanism. Contemporary students will recognize it immediately from smartphones — the challenge is to show them that the same dynamics played out a century earlier.

SUGGESTED DISCUSSION ARC

0–10 min	Open question: 'Does your phone ring in your bedroom? Would your grandmother's have? Does it matter?' Let students respond freely. Surface: the phone extends the threshold of the home.
10–25 min	Brief lecture: Fischer's argument about the 'unintended social use' of the telephone. Telephone companies initially sold it for business use. Women adopted it for sociality. Companies eventually marketed what users had invented. This is a story about who defines technology.
25–50 min	Seminar discussion. Questions 1–3. Question 3 (Goffman) is the most analytically rich — take the most time there.
50–65 min	Graduate section: Marvin, de Certeau, Sennett. Question G1 and G2. The Sennett question connects to the broader public/private theme.
65–75 min	Close: 'The home office is now a standard room in American homes. Fifty years ago it wasn't. Which technology created it, and at what cost?' Brief responses.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Fischer shows that telephone companies initially saw the residential phone as a minor market — it was women's unsanctioned social use that created the market for home telephony. What does that inversion tell us about who actually shapes how technologies develop?**

The gap between the intended and actual use of a technology is a recurring course theme. Mark it explicitly.

2 **Schwartz Cowan argues that labor-saving domestic technologies often increase domestic labor rather than reducing it. Is the telephone a labor-saving device? For whom?**

Gender and labor again — connects to Week 6. Ask students whether the same argument applies to smartphones.

3 **Goffman distinguishes between front-stage (public performance) and backstage (private preparation and relaxation). The telephone brings the audience into the backstage. What does that do to the performance? What does it do to the backstage?**

Productive to ask: where is your backstage right now? Does it still exist?

GRADUATE / MASTER'S QUESTIONS

G1 Marvin's 'When Old Technologies Were New' shows that every new communication technology generates the same social anxieties: it will corrupt youth, undermine face-to-face interaction, erase privacy. Does that historical pattern comfort you or concern you about current anxieties around social media and AI?

G2 Sennett argues in 'The Fall of Public Man' that the erosion of public life leads to narcissism and political passivity. Does the telephone support or undermine his argument? What about the smartphone?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often assume that the home phone was always a domestic technology. Fischer's point is that its domestication was contested and accidental — a different outcome was genuinely possible.

Goffman's dramaturgical metaphor can be taken too literally. Remind students it's an analytical tool, not a claim that people are consciously performing at all times.

Graduate students may want to jump to Sennett's nostalgia critique before engaging seriously with what was actually lost. Hold them in the empirical record before the normative argument.

CONNECTIONS TO OTHER SESSIONS

The front/backstage framework (Goffman) returns in Week 8 (smartphone) and Week 11 (smart home surveillance). The question of 'who discovers the social use of a technology' returns in Week 10 (women's use of micromobility was underestimated by every operator — Hoffman can speak to this). The telephone session is also useful prep for Week 9's discussion of platforms: Uber discovered a use of the smartphone that its creators didn't intend.

FACILITATOR NOTES

> If the class contains students from outside the United States, the American-centric nature of Fischer's history is worth naming. The telephone's spatial consequences differed significantly in high-density apartment cities (New York, Hong Kong, Paris) vs. the suburban single-family home context Fischer describes.

Week 6 · The PC & Domestic Labor

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

The personal computer arrived in the home carrying promises of liberation — from drudge work, from office hierarchy, from limitations of geography. The reality was more complicated: the home computer created new forms of domestic labor (maintenance, troubleshooting, data management), reinforced existing gendered divisions of domestic responsibility, and began the long dissolution of the boundary between work time and leisure time that the smartphone completed. Turkle's ethnographic account of early computer users is vivid and accessible; Wajcman's feminist critique provides the political edge.

THEORETICAL FRAME

The tension between technological promise and domestic reality is the session's engine. Schwartz Cowan's 'consumption junction' concept (introduced last week) returns here with full force: the household is where technology meets daily life, and it is there — not in the lab or the factory — that we see what technology actually does.

SUGGESTED DISCUSSION ARC

0–10 min	Ask: 'Where is your computer right now? Where was your parents' computer when you were growing up? Where was your grandparents' computer, if they had one?' Map the migration of the computing device through domestic space.
10–20 min	Brief lecture: the trajectory of the personal computer from dedicated study to kitchen counter to lap. Each location change is a social negotiation, not just a hardware development. Connect to Week 2's 'hearth sequence.'
20–45 min	Seminar on Turkle and Wajcman. Questions 1–3. Question 2 (who troubleshoots?) usually produces immediate personal recognition and heated discussion.
45–60 min	Graduate section: Haraway's Cyborg Manifesto. Question G1 — this is deliberately provocative. Question G2 on the 'consumption junction.'
60–75 min	Synthesis: 'The laptop is now so lightweight it has no fixed place in the home. Is that liberation or the end of the backstage?' Brief discussion. Preview Week 7 midterm expectations.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Turkle describes early computer users as finding in computing a 'second self' — a mirror for their psychological states and intellectual processes. Is that still how we relate to our devices? What's changed?**

Students born after the smartphone may find this question strange. That's productive — ask them what relationship they do have with their devices.

2 **In most households, who sets up the router? Who installs software updates? Who troubleshoots when the printer doesn't work? What does the distribution of that labor tell us?**

Wajcman's point in practice. Don't let students dismiss this as anecdotal — ask for patterns, not just personal stories.

3 **The personal computer was promised as a tool for liberation from office hierarchy. Did it deliver? Who got liberated, and from what?**

Connects to Hochschild: the double shift. And to the remote work discussion in Week 12.

GRADUATE / MASTER'S QUESTIONS

G1 Haraway's Cyborg Manifesto ends with the slogan 'I would rather be a cyborg than a goddess.' What is she rejecting with 'goddess,' and what is she embracing with 'cyborg'? Does her argument hold up in the age of AI assistants?

G2 Schwartz Cowan's 'consumption junction' asks us to study technology at the point where it is actually used, not where it is produced. Apply that concept to the laptop in the home. What do you see that a study of Apple's design process would miss?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often conflate the argument that technology creates new domestic labor with the argument that technology is bad. Wajcman and Cowan are making an empirical observation, not a polemic. The labor is real; whether it's worth it is a different question.

Haraway's Cyborg Manifesto is frequently misread as celebrating technology uncritically. It is a socialist feminist argument — the cyborg figure is deployed against purity politics and identity essentialism, not in praise of gadgets.

Graduate students sometimes treat the gendered division of domestic tech labor as a 1980s problem that has been solved. Current data suggests otherwise. Worth having on hand.

CONNECTIONS TO OTHER SESSIONS

The domestic labor question connects directly to Week 12 (remote work — who manages the home office environment?) and Week 11 (smart home — who manages the Alexa household?). The Turkle 'second self' concept recurs in Week 8 (smartphone identity) and Week 11 (ambient AI). Preview the midterm: students should be applying one of the three STS frameworks from Week 3 to a case of their choice.

FACILITATOR NOTES

> Week 6 is the last session before the midterm presentations. Use the final 5 minutes to do a quick proposal check-in: 'Raise your hand if you have your primary source identified.' Identify students who are behind and check in with them individually.

Week 7 · Midterm Presentations

SESSION FORMAT	TIMING	LEVEL
Presentations + critique	90 min	UG + Grad

SESSION OVERVIEW

Students present their research proposals. Undergraduate presentations: 5 minutes + 5 minutes Q&A. Graduate presentations: 8 minutes + 7 minutes Q&A. The goal is not a polished product but a clear argument: a technology, a space or social arrangement it changed, a theoretical frame from the course, and a primary source. The instructor's role is to sharpen arguments, identify gaps, and — crucially — connect proposals to each other across the room.

THEORETICAL FRAME

No new theory this week. The session is applied: students demonstrate their command of Weeks 1–6 by deploying it on a case.

SUGGESTED DISCUSSION ARC

0–5 min	Set expectations: the goal of critique is to make the argument stronger, not to find flaws. Model the feedback form: 'The argument I heard was... The evidence that would most strengthen it is... The theoretical frame that fits best is...'
5–75 min	Presentations. For a class of 15: 5 UG students at 10 min each = 50 min; 5 grad students at 15 min each = 75 min. Adjust for class size. Instructor takes notes on the whiteboard, visible to all.
75–90 min	Instructor synthesizes across presentations: 'I noticed three proposals are working with the telephone; two are working with the automobile. What would it look like to put those conversations in dialogue?'

DISCUSSION QUESTIONS — ALL STUDENTS

1 **[For each presentation] What is the central claim? Can you state it in one sentence?**
Force this. If a student can't do it in one sentence after 5 minutes of presentation, the argument isn't clear yet.

2 **What is the primary source, and how does it give you access to something that secondary sources can't?**
Graduate students especially should be held to this.

3 **Which framework from Week 3 — SCOT, ANT, or domestication theory — maps most**

> **naturally onto your case? What does adopting that frame require you to look at, and what does it require you to ignore?**
> *Good for graduate students especially.*

GRADUATE / MASTER'S QUESTIONS

G1 What is the counterargument your paper will need to address? Who would push back on your claim, and why?

G2 Is your case generalizable? Or is it a single example? If the latter, what does it prove?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often present summaries of readings rather than arguments. Interrupt gently: 'What do you think about that?' or 'What does that mean for your case?'

Graduate students sometimes aim too broad — 'technology and society in the 20th century.' Push for a specific technology, a specific space, a specific transformation.

CONNECTIONS TO OTHER SESSIONS

The proposals are the seeds of the final papers. Document the connections you see across proposals — you'll use them in Week 13 (peer critique) to deliberately pair students whose work speaks to each other.

FACILITATOR NOTES

Have a visible timer. Allow the room to ask questions — the best critique often comes from peers, not the instructor. Write each presenter's central claim on the board and leave it there for the whole session. By the end, the board becomes a map of the course's intellectual landscape.

Week 8 · The Smartphone & Third Places

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

Ray Oldenburg coined 'third place' in 1989 — before the smartphone, before the laptop, before ubiquitous WiFi. His argument (that coffee shops, barbershops, pubs, and parks are essential to democratic civic life) is now read against the backdrop of a technology that simultaneously invaded those places and, in some cases, created new virtual versions of them. The tension between Oldenburg's theory and the smartphone's reality is the engine of this session.

THEORETICAL FRAME

Turkle's 'Alone Together' argues that digital connectivity produces a paradox of presence: we are technically connected to more people than ever but experientially alone in shared physical space. That paradox maps directly onto Oldenburg's concern. The question is whether the third place has migrated, degraded, or merely changed form.

SUGGESTED DISCUSSION ARC

0–10 min	Ask students to list the physical spaces where they spend time that are not home or school/work. How many of those spaces involve smartphone use? How many explicitly discourage it? Have any been designed around it?
10–20 min	Brief presentation of Oldenburg's criteria for a third place: neutral ground, leveler, conversation is the main activity, accessible, playful, a home away from home. Ask: does the coffee shop you work in meet these criteria? Does your favorite corner of the internet?
20–50 min	Seminar discussion. Questions 1–3. Question 3 (has the third place moved online?) is the most contested — let it run.
50–65 min	Graduate section: Wellman, Hampton, Lofland. Question G1 and G2.
65–75 min	Close: 'If you could design a space that served as a genuine third place in 2026, what would it look like? What would the technology policy be?' Quick sketches or verbal descriptions.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Oldenburg's third places — pubs, barbershops, coffee shops — emerged in specific economic and cultural contexts. Are those contexts reproducible? Or are we trying to**

preserve something that was always contingent?
Keep students from nostalgizing. Ask: who was included in those third places? Who was excluded?

2 **Turkle argues that smartphones allow us to be 'alone together' — present in body but absent in attention. Is that a new phenomenon, or have there always been ways to be present and absent simultaneously?**
Students often bring up books, newspapers, daydreaming. Good. Ask what's different about the smartphone specifically.

3 **Has the third place moved online? If Discord servers, Reddit communities, and Slack workspaces perform the social functions Oldenburg describes, does the physical location matter?**
The hardest question of the session. There's no clean answer — that's the point.

GRADUATE / MASTER'S QUESTIONS

G1 Wellman's 'networked individualism' argues that social support has shifted from place-based groups (neighborhoods, parishes) to personal networks maintained across space. Is that a loss? Does it depend on who you are?

G2 Lofland distinguishes between parochial, public, and private realms. Where does the smartphone sit in that taxonomy? Does a coffee shop with WiFi constitute a public, parochial, or private realm?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often read Turkle as simply anti-technology. She is making an empirical and psychological argument about connection and loneliness — not a moral argument about screen time. Keep the analysis structural.

The 'third place has moved online' argument is seductive but needs pressure. Oldenburg's criteria include physical co-presence, leveling of status, and accessibility without payment. Online spaces often fail at least two of these.

Graduate students sometimes dismiss Oldenburg as sociologically naive. His criteria are normative and idealized — acknowledge that, but ask whether the normative standard is useful precisely because it's demanding.

CONNECTIONS TO OTHER SESSIONS

Third places return in Week 12 (remote work — the coffee shop as de facto office) and Week 9 (platform urbanism — Starbucks as a platform for remote work, Airbnb as a privatized version of the hotel lobby as third place). Turkle's 'alone together' concept is the precursor to the surveillance capitalism argument in Week 11.

FACILITATOR NOTES

> The design exercise at the end of the session works very well in studios with physical making capacity — students can quickly sketch a space. In a seminar without that capacity, a verbal description works fine. The constraint ('what is the technology policy?') is what makes it interesting.

Week 9 · Platform Urbanism

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

Platform capitalism (Srnicek) describes a new form of business logic in which value is extracted from the interaction between two or more groups, facilitated by a digital infrastructure. When that logic is applied to urban space — housing, transportation, food delivery — the consequences are spatial as well as economic. Airbnb doesn't just disrupt hotels; it removes apartments from the long-term rental market and restructures the residential use of city blocks. Uber doesn't just compete with taxis; it redefines the purpose and management of the curb.

THEORETICAL FRAME

Wachsmuth and Weisler's 'Airbnb and the Rent Gap' is a rigorously spatial argument: platforms don't just respond to urban conditions, they actively produce new spatial configurations. This is the strongest empirical reading of the semester and should be treated with corresponding analytical care.

SUGGESTED DISCUSSION ARC

0–10 min	Ask: 'Has anyone in this room stayed in an Airbnb? Taken an Uber? Had food delivered in the last 48 hours?' Virtually every hand goes up. 'You have participated in platform urbanism. What did you do to the city?'
10–20 min	Brief lecture: what makes a business a 'platform'? Three characteristics: it mediates between two or more groups; it benefits from network effects; it extracts data as a byproduct of every transaction. Urban platforms apply this logic to housing, mobility, and food.
20–50 min	Seminar discussion. Questions 1–3. The Airbnb question (Question 1) is most productive for urban planning students; the Uber question (Question 2) for mobility students. Run both.
50–65 min	Graduate section: Zuboff and Rosenblat. Question G1 and G2.
65–75 min	Close: 'If you were a city council member, what would you regulate about Airbnb? What tools do you have? What tools are you missing?' Brief policy exercise.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Wachsmuth and Weisler argue that Airbnb doesn't respond to the rent gap — it creates it. Walk me through that mechanism. How does a platform application shift the value**

extraction logic of an entire housing market?
This is a causal chain question. Students need to work through: platform removes units from rental stock → supply decreases → rents rise → rent gap widens → more units converted to Airbnb. Each step matters.

2 **Uber and Lyft have produced enormous amounts of congestion in dense urban cores — more vehicle miles traveled, not fewer. How is that possible for a 'ride-sharing' platform? Who absorbs the cost?**
The externality question. Platforms internalize profit and externalize costs — onto streets, onto drivers, onto air quality.

3 **Srnicek argues that platforms accumulate data as a form of capital. What data does Airbnb accumulate? What does it do with it? Who else would want it?**
Privacy, power, and urban intelligence intersect here. Connects forward to Week 11.

GRADUATE / MASTER'S QUESTIONS

G1 Zuboff's 'surveillance capitalism' describes a logic in which human experience is the raw material for prediction products sold to advertisers and others. Apply that logic to Airbnb. Who is being predicted, and for whom?

G2 Rosenblat argues that Uber's algorithmic management creates a new form of labor control that is invisible precisely because it is automated. What spatial consequences does invisible labor control produce? Think about where drivers wait, how they navigate, what neighborhoods they serve.

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often frame platforms as neutral facilitators of exchange. Srnicek's point is that they are not neutral — they structure what exchanges are possible, who can participate, and on what terms.

The Airbnb critique is sometimes heard as anti-tourism or pro-regulation per se. Keep it spatial: the argument is about what happens to housing stock and neighborhood character, not about whether tourism is good.

Graduate students may treat Zuboff as too totalizing. Acknowledge the critique (she is sometimes accused of conspiracy thinking) while taking seriously the empirical claims about data extraction.

CONNECTIONS TO OTHER SESSIONS

Platform urbanism is the contemporary form of the 'policy shapes space' argument from Week 4 (automobile). The question 'who owns the curb?' connects Week 4 (parking minimums) to Week 9 (Uber) to Week 10 (dockless bikes). In Week 11, the smart home is revealed as a platform that extracts data from domestic space.

FACILITATOR NOTES

> The policy exercise at the end works best if students have looked up their city's short-term rental regulations before class. Assign this as a 15-minute pre-class task. The contrast between cities with strong regulations (New York, Barcelona, Amsterdam) and those without is immediately instructive.

Week 10 · Micromobility & Street Design

SESSION FORMAT	TIMING	LEVEL
Seminar + guest practitioner	90 min	UG + Grad

SESSION OVERVIEW

This is the session where the course's theoretical work meets practitioner reality. Patrick Hoffman's account of building Social Bicycles (SoBi) and the product that became JUMP is a first-hand account of how a mobility technology interacts with urban space, regulatory environments, and social equity. The NACTO guide grounds the design dimensions; Sheller's Mobility Justice reframes the whole conversation in terms of access and power. If Patrick is available to guest, this session is one of the most memorable of the semester.

THEORETICAL FRAME

The central tension is between mobility as design problem (NACTO) and mobility as justice problem (Sheller). Those are not the same framing, and they lead to different interventions. The practitioner account complicates both: building a product that works in cities requires engaging with both simultaneously, usually under commercial pressure.

SUGGESTED DISCUSSION ARC

0–10 min	Open with a question: 'Has anyone here used a bike share or scooter share system? What was the experience like? What was good? What failed?' Surface: coverage gaps, parking conflicts, equity of access, reliability.
10–25 min	If Patrick is guest: structured interview format. Suggested opening questions below. If no guest: brief lecture on the SoBi/JUMP arc — what dockless bikes required cities to decide, and what that decision-making process revealed about urban politics.
25–50 min	Seminar discussion. Questions 1–3. Question 2 (equity) is where the richest disagreement usually lives.
50–70 min	Graduate section: Sheller, Lydon, SFMTA pilot report. Question G1 and G2. The SFMTA report is a primary policy document — treat it as such. What does it reveal about how cities make decisions?
70–90 min	Close: 'What would a truly just micromobility system look like? What would it require from the city, from the operator, from the user?' Small groups, then share back.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Patrick Hoffman's account of dockless bikes shows that the technology itself was relatively simple — the hard problems were regulatory, spatial, and political. What does that tell us about what kind of expertise matters most for building products in cities?**

Urban expertise, not just engineering expertise. The question is: who should be in the room when these decisions are made?

2 **Bikeshare systems in the United States have consistently underserved low-income and minority neighborhoods. Is that a design problem, a business model problem, or a political problem?**

The answer is: all three, interacting. The goal is to trace the interactions, not pick one.

3 **The NACTO Urban Street Design Guide represents a professional consensus about how streets should work. Who participated in creating that consensus? Who didn't? How does that shape what the guide recommends?**

Standard-setting as a political act. Connects to Week 4 (zoning codes as political documents).

GRADUATE / MASTER'S QUESTIONS

G1 Sheller's 'Mobility Justice' argues that the right to move through space is unequally distributed along lines of race, class, gender, and disability. Apply that framework to dockless scooters in [city where course is taught]. Who can use them? Who cannot? What would justice require?

G2 The SFMTA scooter pilot report is a government document. What interests does it represent? What does it measure that tells you what the city values? What does it not measure, and why might that be?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes assume that adding a micromobility option automatically improves equity. Sheller's point is that mobility justice requires more than more modes — it requires addressing the unequal spatial distribution of origins and destinations, not just vehicles.

The 'dockless' innovation is often understood as purely technical (no station needed). Its deeper innovation was regulatory: it forced cities to make explicit decisions about public space that had previously been unspoken. Make sure students grasp this.

Graduate students may be skeptical of practitioner accounts as theoretically unrigorous. Push back: the practitioner account gives access to contingency, trade-off, and failure that theoretical accounts rarely capture.

CONNECTIONS TO OTHER SESSIONS

Micromobility is the bridge between the automobile session (Week 4), the platform session (Week 9), and the street design future (Week 14). The 'who owns the curb?' question runs through all three. If Patrick guest-teaches, his account of GPS-enabled fleet management connects forward to Week 11's surveillance capitalism discussion.

FACILITATOR NOTES

> Guest practitioner format: prepare Patrick (or any guest) with 3–4 specific questions in advance. The best format is a structured conversation, not a lecture — students can ask follow-ups after each exchange. Suggested opening questions for Patrick: (1) 'What was the moment you realized the product was a city problem, not just a technology problem?' (2) 'What did cities get wrong when they tried to regulate dockless bikes?' (3) 'What would you do differently?'

Week 11 · AI & the Intelligent Home

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

The smart home represents the logical culmination of the course's domestic technology arc: from the television that rearranged the living room (Week 2) to the smartphone that collapsed public and private (Week 8) to the ambient AI system that converts the home itself into a data extraction platform. Zuboff's surveillance capitalism framework and Mattern's smart city critique are the theoretical tools; Crawford's material account of AI grounds the abstract in physical infrastructure.

THEORETICAL FRAME

The key move in this session is from 'smart home as convenience' to 'smart home as platform.' Every Alexa query, every Nest thermostat adjustment, every Ring doorbell recording is both a service and a data extraction event. Zuboff's argument: the service is the pretext; the data is the product. Students need to work through whether that framing is paranoid or merely accurate.

SUGGESTED DISCUSSION ARC

0–10 min	Ask: 'Who has a smart speaker, smart thermostat, or video doorbell at home?' Count hands. 'Who read the terms of service before setting it up?' (Usually zero or one.) 'What did you agree to?'
10–20 min	Brief lecture: the three layers of the smart home. Layer 1: the device (what it does for you). Layer 2: the platform (what data it collects and for whom). Layer 3: the infrastructure (where that data goes, who runs the data centers, at what cost to the environment). Crawford's 'Atlas of AI' is most useful here.
20–50 min	Seminar discussion. Questions 1–3. Question 2 (prediction vs. service) is the conceptual core — take the most time there.
50–65 min	Graduate section: Zuboff Ch. 1–2, Rose, Lupton. Question G1 and G2.
65–75 min	Close: 'If you designed a smart home system that was genuinely in the user's interest rather than the platform's, what would it look like? What business model would support it?' Brief speculation — intentionally open-ended.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Mattern argues that calling a city 'smart' usually means it has been made legible to computational systems — not that it has actually become wiser or more responsive to its**

residents. Does that critique apply to the 'smart home' as well?

The analogy is tight: just as smart city dashboards optimize for measurable metrics rather than lived quality, smart home systems optimize for user engagement and data yield rather than genuine domestic wellbeing.

2 **Zuboff distinguishes between a product that serves the user and a product that extracts behavioral data from the user as a form of raw material. Is your smart speaker a product or an extraction device? Can it be both?**

The honest answer is: it's both simultaneously. The question is whether that's a problem, and for whom.

3 **The Ring doorbell creates a private surveillance network around the home. What does it do to the relationship between neighbors? Between residents and police? Between the domestic and the public?**

Connects Weeks 5 and 9: the threshold question returns, now with cameras.

GRADUATE / MASTER'S QUESTIONS

G1 Rose's 'Governing the Soul' argues that contemporary power operates not through external coercion but through the internalization of norms — we govern ourselves. How does the smart home enact that form of power? Who is governing whom?

G2 Crawford's 'Atlas of AI' traces the physical supply chains of artificial intelligence — the rare earth mines, the data centers, the gig workers labeling training data. How does that material account change your analysis of the 'intelligence' of the smart home?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often frame smart home critique as privacy paranoia. Zuboff's argument is about power, not just privacy: the issue isn't that someone can hear your conversations, it's that your behavioral data is used to predict and modify your future behavior without your knowledge.

The convenience/surveillance trade-off is often presented as a free choice. Push back: the choice is structured by defaults, by social pressure, and by the unavailability of alternatives that don't extract data.

Graduate students may treat Zuboff as too binary (serving users vs. extracting data). Acknowledge the critique — the framework is deliberately polemical — while taking seriously the empirical documentation.

CONNECTIONS TO OTHER SESSIONS

This session is the apex of the domestic technology arc (Weeks 2, 5, 6, 8, 11). It also sets up Week 12: the home office in 2026 is a smart home subsystem, with corporate monitoring software sitting alongside Alexa in the same domestic space. And it connects back to Week 9: the smart home is a platform, and like Airbnb, it extracts value from domestic space.

FACILITATOR NOTES

> The terms-of-service opening works reliably. If you want to extend it: pull up the Amazon Echo privacy policy on screen and spend 3 minutes walking through what it actually says. The language is designed to be opaque — that opacity is itself a data point. For sessions with architecture students, the question 'what would a smart home look like if it were designed for residents rather than platforms?' tends to generate rich spatial proposals.

Week 12 · Remote Work & the Office

SESSION FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

The pandemic made remote work a mass phenomenon, accelerating a shift that was already underway. The spatial consequences — for downtowns, for transit systems, for home square footage, for the geography of economic activity — are still unfolding. This session places those consequences in historical context (the office has always been a designed artifact that embeds power relations, as Saval shows) and in theoretical context (Sennett's ethics of dwelling, Wajcman's time pressure). The goal is for students to see the remote work 'debate' as a spatial problem, not just a management preference.

THEORETICAL FRAME

The session holds two arguments in tension: Glaeser's pro-density argument (cities are engines of productivity and innovation, and remote work degrades the agglomeration effects that make them work) vs. the feminist and critical tradition (the office was always a tool of gendered and racial hierarchy, and its disruption opens space for more equitable arrangements). Neither is simply right.

SUGGESTED DISCUSSION ARC

0–10 min	Ask: 'Where do you work best? Has the answer to that question changed since you started this program? Why?' Surface: the question of 'best' is doing a lot of work — best for what? For whom?
10–20 min	Brief lecture: the history of the office as a designed environment. Saval's arc: the Victorian counting house → the open-plan bullpen → the cubicle farm → the open office → the Zoom room. Each configuration embeds assumptions about labor, supervision, and collaboration.
20–50 min	Seminar discussion. Questions 1–3. Question 2 (who does remote work serve?) generates the most productive disagreement.
50–65 min	Graduate section: Useem's Atlantic piece (current), Glaeser, Wajcman. Question G1 and G2.
65–75 min	Close: 'What does your ideal workplace look like, spatially? What technology does it require? What does it do to the city around it?' Brief responses — connecting personal preference to urban consequence.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Saval argues that the office's physical design has always reflected and reinforced organizational hierarchies — who gets walls, who gets windows, who faces the supervisor. What did the open-plan office change, and what did it leave in place?**

The open plan was sold as democratic and collaborative but often increased surveillance and reduced privacy. Ask students to apply Winner's 'artifacts have politics' to the open-plan office.

2 **Remote work was celebrated as liberation from the office and simultaneously criticized as the invasion of work into the home. Both of those things are true. Who experiences each, and does it depend on your job, your home, your household composition?**

The distribution of remote work's benefits and costs is deeply unequal. Push students to specify whose experience they're describing.

3 **If large numbers of knowledge workers stop commuting to downtown offices, what happens to the city? To transit ridership? To the small businesses that serve commuters? To property values? Trace the cascade.**

Urban systems thinking — this is the Week 4 'automobile shapes city' argument in reverse.

GRADUATE / MASTER'S QUESTIONS

G1 Glaeser argues that cities derive their productivity advantage from density — the spontaneous collisions of ideas and people that happen in corridors and coffee shops. Is that argument testable? What would falsify it? Does the remote work experiment constitute a test?

G2 Wajcman's 'Pressed for Time' argues that technology promises time liberation but consistently delivers time pressure. Apply that argument to remote work and the calendar management tools that came with it. Who gained time? Who lost it?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often treat remote work as an individual preference question ('some people are more productive at home'). The course argument is that it's a spatial and political question: remote work restructures cities, real estate markets, transit systems, and domestic arrangements.

The 'office is over' and 'office is back' narratives are both too simple. The interesting question is: which workers, in which roles, in which cities, under which management regimes? Specificity is everything.

Graduate students sometimes treat Glaeser as a conservative apologist for real estate capital. His agglomeration argument has empirical support and deserves engagement, even if his policy conclusions are contestable.

CONNECTIONS TO OTHER SESSIONS

Remote work is the contemporary form of the home/work blurring that began with the PC (Week 6) and deepened with the smartphone (Week 8). Glaeser's agglomeration argument connects back to Week 4 (why cities exist — the same reason suburbia undermined them). The smart home (Week 11) is now also the smart office — the two spaces have merged.

FACILITATOR NOTES

> If the course is taught in a city with a significant downtown office market, bring current vacancy rate data. The contrast between, say, San Francisco's ~30% office vacancy and a mid-size Midwestern city's relative stability is instructive. Students from different geographies often have very different intuitions about this, and the contrast is productive.

Week 13 · Studio Critique I

SESSION FORMAT	TIMING	LEVEL
Peer critique workshop	90 min	UG + Grad

SESSION OVERVIEW

The peer critique session is structured around submitted drafts. Each student reads and prepares written feedback on two assigned papers before class. The session is a workshop: not a series of presentations, but a conversation around specific passages, arguments, and evidence. The instructor's role is to moderate, elevate the conversation, and model the kind of critical engagement they want students to practice.

THEORETICAL FRAME

The readings (Graff & Birkenstein, Sword, Hayot) are writing-craft texts. They should be assigned not as readings to discuss, but as tools to use: students bring them to the session and apply them to the papers in front of them.

SUGGESTED DISCUSSION ARC

0–10 min	Set the frame: 'Today we are here to make arguments stronger, not to evaluate whether they're good enough. The question for every piece of feedback is: what would make this argument more convincing to a skeptical reader?'
10–60 min	Workshop pairs (UG) or triads (Grad). Each group spends 20–25 minutes on each paper: 5 minutes of quiet reading/annotation, then 15 minutes of structured conversation. Instructor circulates.
60–80 min	Plenary: each group shares one observation — 'The strongest argument in the papers we read was... The most common gap was...' Instructor synthesizes across groups.
80–90 min	Instructor closes with 3–5 specific observations about the cohort's writing at this stage. Not about individual papers — about patterns.

DISCUSSION QUESTIONS — ALL STUDENTS

1. **What is the paper's central claim? Can you state it more sharply than the author did?**
   Graff & Birkenstein's 'they say / I say' structure: what is the author responding to, and what are they saying in response?

2. **What is the best piece of evidence in this draft? What evidence is missing that would make the argument airtight?**

Evidence gaps are usually the most actionable feedback.

3 **Where does the paper use a theoretical framework, and where does it drop it? Is the framework doing analytic work, or is it decorative?**

The most common graduate student problem: ANT or SCOT is introduced and then abandoned when the actual analysis begins.

GRADUATE / MASTER'S QUESTIONS

G1 Apply Hayot's 'Uneven U' structure to this paper. Where is the paper on that arc? Is it developing the argument or restating it? Where should it be heading in the next draft?

G2 What is the paper's relationship to its primary source? Is the author using the source to build an argument, or using the argument to introduce the source?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often give feedback that is too gentle to be useful. Model direct, specific, constructive critique: 'On page 3, the argument assumes X without establishing it. Here's how I'd establish it.'

Some students give feedback that is too abstract: 'The argument could be clearer.' Push for specificity: 'The argument in paragraph 2 becomes unclear when you introduce Latour without explaining how ANT applies to this case.'

CONNECTIONS TO OTHER SESSIONS

The specific arguments being critiqued are the final paper arguments. Use this session to identify the two or three students who need the most support and schedule individual meetings before Week 15.

FACILITATOR NOTES

Pair students strategically: put proposals that are in dialogue with each other in the same critique group. The best critique often comes from someone working on an adjacent problem who asks 'but what about X?' and X is exactly what the paper needs.

Week 14 · Designing the Response

SESSION FORMAT	TIMING	LEVEL
Seminar + design exercise	75 min	UG + Grad

SESSION OVERVIEW

The penultimate session asks the question that the course has been building toward: if technologies reshape spaces in ways that often create new social problems, what can architects, planners, and product designers do about it? This is not a session about solutions — it is a session about design posture. Brand's 'How Buildings Learn' offers the most useful frame: buildings (and cities, and products) that can adapt to change over time are more resilient than those optimized for a single use case. The challenge is to apply that insight to the present moment of AI-driven spatial transformation.

THEORETICAL FRAME

The tension between formal ambition (Aureli, Koolhaas) and pragmatic adaptation (Brand, Duany et al.) is the design-theory version of the course's central tension. Brand asks: how do you design for change you can't predict? Koolhaas asks: is the ambition to design for resilience itself a surrender to capital? Till mediates: architecture is always incomplete, always dependent on conditions it doesn't control — and that's not a failure, it's the condition of practice.

SUGGESTED DISCUSSION ARC

0–15 min	Open with Brand's 'shearing layers' diagram: site, structure, skin, services, space plan, stuff — each changes at a different rate. Ask: 'Where does a technology intervention land on this diagram? And what does that tell you about how long its spatial consequences last?'
15–35 min	Seminar discussion. Questions 1–3. Question 2 (Co-op City) connects to Hoffman's @arcktip essay and is a good place to anchor the theoretical discussion in a specific built case.
35–55 min	Design exercise: small groups each receive a different brief. Brief A: redesign a downtown office building for a post-remote-work era. Brief B: design a neighborhood-scale micromobility hub that is also a third place. Brief C: design a public housing typology that accommodates changing domestic technology without requiring structural renovation. 15 minutes of sketching/discussion, then share.
55–70 min	Graduate section: Koolhaas, Till, Aureli. Question G1 and G2.
70–75 min	Close: 'What does this course have taught you that you will carry into your practice?' Brief, personal, no wrong answers. Preview Week 15 expectations.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **Brand argues that buildings that learn — that can adapt to new uses, new technologies, new inhabitants — are more valuable and more humane than buildings designed for a specific optimized program. Does that argument scale to cities? To products?**

The Amazon fulfillment center is optimized for one use. The New York City street grid is not optimized for anything — and that's why it's still useful 200 years later. Which approach is more resilient?

2 **Co-op City in the Bronx was designed in the late 1960s as a comprehensive response to the housing crisis — massive scale, full services, designed community. Hoffman's @arcktip essay asks what we can learn from it. What can we? What are its limits as a model?**

This is a normative question about what good design at scale looks like. There's no clean answer.

3 **Duany, Plater-Zyberk, and Speck argue that the principles of traditional urbanism — mixed use, walkability, a clear hierarchy of streets — are the most resilient design response to technological change. Is that conservatism, or is it empirically right?**

The new urbanist position is genuinely contested. Push students to make arguments, not just take sides.

GRADUATE / MASTER'S QUESTIONS

G1 Koolhaas's 'Junkspace' argues that contemporary architecture is dominated by the logic of capital — interchangeable, placeless, optimized for consumption rather than habitation. Is that a critique or a description? And what, if anything, can architects do about it?

G2 Till's 'Architecture Depends' argues that architecture's dependence on contingent social conditions is not a failure to be overcome but the very condition of its ethical practice. How does that position relate to the course's argument about technology and space?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read Brand's adaptability argument as 'design should be neutral or flexible.' That's not quite right — Brand is arguing for layers of intentionality, not the absence of intention. The site and structure carry long-term commitments; the stuff layer changes freely.

The new urbanist position (Duany et al.) is sometimes dismissed as nostalgic. Worth noting: the empirical evidence on walkability and mixed use as correlates of resilience is reasonably strong. The nostalgia critique applies to aesthetic claims, not necessarily to planning principles.

Graduate students sometimes treat Koolhaas and Till as incompatible. They're working on different scales and in different modes — Koolhaas is diagnostic and provocative; Till is ethical and practical. They can coexist.

CONNECTIONS TO OTHER SESSIONS

This session is the payoff for the whole course. Every case study from Weeks 2–12 should be available to pull forward: what would a space designed with that lesson in mind look like? The design exercise is the applied form of the course's central question. Return explicitly to Week 1: 'What would you say now to the person who arranged the furniture around the television?'

FACILITATOR NOTES

> The design briefs work best when they are genuinely constrained. Give groups a specific site (a real building, a real block in the course's city), a specific technology transition to respond to, and a specific user group. Open-ended briefs produce generic responses. Tight briefs produce interesting ones.

Week 15 · Final Presentations

SESSION FORMAT	TIMING	LEVEL
Public presentations + external critics	120 min	UG + Grad

SESSION OVERVIEW

The final session is a public exhibition of completed work. Undergraduate presentations are 10 minutes + 5 minutes Q&A. Graduate presentations are 15 minutes + 10 minutes Q&A. Invite 2–3 external critics from architecture, urban planning, or technology practice. The instructor's role is to frame the session, moderate critique, and draw the intellectual threads of the course together in closing remarks.

THEORETICAL FRAME

No new theory. The session tests whether students can apply the course's full toolkit — the STS frameworks, the historical cases, the spatial analysis — to a new problem of their own choosing.

SUGGESTED DISCUSSION ARC

0–5 min	Brief framing: 'We started this course with a question: how do technologies reshape the spaces we inhabit? Today you answer it with your own evidence.'
5–90 min	Presentations. For a class of 15: 5 UG at 15 min = 75 min; 5 grad at 25 min = 125 min. Adjust to class size. Leave 15–20 min for final synthesis.
90–110 min	External critics offer brief observations (5 min each). Instructor moderates a plenary conversation: 'What surprised you across these presentations? What questions are still open?'
110–120 min	Instructor closing: return to the living room image from Week 1. 'Here is what this cohort said about it, in 15 different registers.' A brief, generous, honest synthesis of what the class learned together.

DISCUSSION QUESTIONS — ALL STUDENTS

1 **[For each presentation] What is the central claim, and what is the most surprising piece of evidence that supports it?**
Push for surprise. The best research finds something that complicates the obvious story.

2 **What STS framework is doing the most analytical work in this paper? Where does the framework's limitations show up?**
Graduate students especially should demonstrate self-awareness about their analytical tools.

3 **What does this case teach us that we couldn't have learned from any other? Why this technology, this space, this moment?**
The specificity of the case is its value. Honor that.

GRADUATE / MASTER'S QUESTIONS

G1 What is the generative tension in this paper — the place where the argument could have gone either way? How did you resolve it, and are you satisfied with the resolution?

G2 If you were teaching this course next year, which session would you add this paper to, and what would you ask students to discuss?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes present conclusions rather than arguments. Remind them: the presentation should show the thinking, not just the finding.

External critics sometimes reframe student work in their own terms. That's valid, but protect the student's framing first — ask the student to respond before the conversation moves on.

CONNECTIONS TO OTHER SESSIONS

The final session closes the loop opened in Week 1. The course question — 'how do technologies reshape the spaces we inhabit?' — has been answered 15 different times, in 15 different ways, by 15 different investigators. That is the course's argument.

FACILITATOR NOTES

External critics: brief them in advance. Share one or two student abstracts. Ask them to come prepared with one question for the cohort, not just for individual papers. The best final sessions feel like a conversation the whole class is having, not a series of one-on-one evaluations. If Patrick Hoffman is willing to serve as an external critic for final presentations, this creates a meaningful bookend: the practitioner who anchored Week 10 now responds to the full range of student inquiry.

APPENDIX: QUICK-REFERENCE QUESTION BANK BY THEME

The following questions can be deployed across multiple sessions whenever discussion needs redirection or deepening:

Technology & Space (use in any session)

– What specific spatial arrangement did this technology require, enable, or foreclose?
– Who paid for the space that this technology created? Who was excluded from it?
– If you removed this technology from the space, what would you see?

Power & Politics (Winner / Zuboff / Sheller)

– Who made the key decisions, and who was not in the room?
– Who bears the externalized costs of this technology?
– If this technology has politics, who benefits from those politics?

Gender & Labor (Wajcman / Cowan / Hochschild)

– Who manages this technology in the household? Who troubleshoots it?
– Did this technology reduce labor or redistribute it?
– Whose time was saved, and whose was not?

Theory Application (STS frameworks)

– Apply SCOT: who are the relevant social groups, and what does this technology mean to each?
– Apply ANT: what human and nonhuman actors need to be enrolled for this technology to work?
– Apply domestication theory: where is this technology in the appropriation → conversion sequence?

Practitioner vs. Academic (Weeks 10, 14)

– What can Patrick's account of building JUMP see that academic papers on bikeshare cannot? What does it miss?
– What would have to be true for a designer to have prevented the spatial problem we're discussing?
– If you were advising a city council on this technology, what would you say?

This guide is a living document. Instructors who use it are encouraged to annotate, revise, and return observations to Patrick Hoffman (@arcktip) so that future iterations can benefit from classroom experience across institutions.